

SOP-PS-210: Pressure Relief Valve Testing and Set-Pressure Verification

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Section 1. Purpose and Scope

This procedure governs the bench testing, set-pressure verification, and return-to-service of spring-loaded pressure relief valves (PRVs) protecting overpressure-critical equipment. It applies to all PRVs listed on the Pressure Relief Device Register.

Section 2. Test Intervals

Clause 2.1 — Every PRV in overpressure-critical service shall be removed, bench tested, and set-pressure verified at each unit turnaround interval, not to exceed 60 months, whichever is sooner.

Clause 2.2 — A PRV protecting a fouling or corrosive service shall be tested at the shorter interval specified on its device datasheet, regardless of the general turnaround schedule.

Section 3. Set-Pressure Tolerance and Unscheduled Testing

Clause 3.5 — The as-found set pressure at bench test must fall within $\pm 3\%$ of the nameplate set point. A result within tolerance with no observed seat leakage at 90% of set pressure confirms the valve is fit for continued service without adjustment.

Clause 3.7 — Any recorded upstream pressure excursion exceeding 105% of the protected system's design pressure triggers an unscheduled bench test of that system's PRV before the next routine interval, regardless of how recently it was last tested.

Clause 3.8 — An as-found result outside the $\pm 3\%$ tolerance band requires valve removal, re-set to nameplate, and a documented root-cause review before return to service. The root-cause review shall determine whether the drift indicates a service-specific fouling/corrosion issue requiring a shortened future test interval under Clause 2.2.

Section 4. Installation and Documentation

Clause 4.1 — A re-set or replacement PRV shall not be reinstalled until its bench test certificate is signed by a qualified technician and attached to the device's permanent record.

Clause 4.3 — Nameplate set pressure, as-found pressure, and as-left pressure shall all be recorded on the test certificate; any deviation between as-found and nameplate that falls inside tolerance is still recorded, not just deviations that trigger Clause 3.8.

Section 5. Records

Clause 5.2 — Every completed test certificate is filed to the Pressure Relief Device File. Retain until the next scheduled test interval plus 5 years.